

ABSTRACT

Recent advancements in large language models (LLMs) have significantly enhanced the capabilities of collaborative multi-agent systems, enabling them to address complex challenges. However, within these multi-agent systems, the susceptibility of agents to collective cognitive biases remains an underexplored issue. A compelling example is the Mandela effect, a phenomenon where groups collectively misremember past events as a result of false details reinforced through social influence and internalized misinformation. This vulnerability limits our understanding of memory bias in multi-agent systems and raises ethical concerns about the potential spread of misinformation. In this paper, we conduct a comprehensive study on the Mandela effect in LLM-based multi-agent systems, focusing on its existence, causing factors, and mitigation strategies. We propose MANBENCH, a novel benchmark designed to evaluate agent behaviors across four common task types that are susceptible to the Mandela effect, using five interaction protocols that vary in agent roles and memory timescales. We evaluate agents powered by several LLMs on MANBENCH to quantify the Mandela effect and analyze how different factors affect it. Moreover, we propose strategies to mitigate this effect, including prompt-level defenses (*e.g.*, cognitive anchoring and source scrutiny) and model-level alignment-based defense, achieving an average 74.40% reduction in the Mandela effect compared to the baseline. Our findings provide valuable insights for developing more resilient and ethically aligned collaborative multi-agent systems¹.

1 INTRODUCTION

“Memory is deceptive because it is colored by today’s events.” — Albert Einstein

With the widespread deployment of large language models (LLMs) (Kojima et al., 2022), LLM-based multi-agent systems (Li et al., 2023a; Hong et al., 2024; Wu et al., 2025c) are increasingly used to address complex problems in fields like public policy analysis and social governance (Zhang et al., 2024; Liu et al., 2026). A key strength of these systems is their ability to simulate social dynamics like deliberation and consensus-building. However, this very capability introduces a significant risk: the emergence of **collective cognitive biases** analogous to those in human groups. For instance, the **Mandela effect** (Prasad & Bainbridge, 2022) is a well-known human cognitive bias (Liu et al., 2025a) where a group shares a false memory of a verifiable fact². Just as this effect can arise from memory fragility and social reinforcement in people, LLM agents interacting within a system could develop similar distortions, leading to flawed group judgments. This could impact their collective problem-solving abilities or even raise significant ethical concerns. For example, the replication of

collective cognitive biases might lead to shared false memories spreading through LLM interactions, causing agents to ignore the truth when influenced by suspicious evidence. Such effect undermines the reliability of AI-driven information systems and increases misinformation risks in high-stakes scenarios, including contract review (Narendra et al., 2024), and fact-checking (Tang et al., 2024).

While existing research has explored related topics, including LLM hallucination (Huang et al., 2025) and multi-agent debate (Liang et al., 2024), a critical gap remains in understanding the Mandela effect (*i.e.*, collective false memory). This gap arises from several limitations in prior work. First, prior work mainly focuses on individual agent errors (Xu et al., 2024) or simple conformity (Zhu et al., 2025; Weng et al., 2025), ignoring the unique aspects of the Mandela effect (*e.g.*, persuasive specious evidence in agent interactions that spread shared false memories throughout a system). Second, hallucination is often viewed as a stateless, one-shot failure, which overlooks the memory-related nature of the Mandela effect, involving the consolidation and persistent recall of a socially shared falsehoods. Moreover, no current benchmark systematically evaluates this phenomenon. To fill this gap, it is crucial to understand how multi-agent systems replicate human cognitive biases, and how to measure and mitigate the Mandela effect, which is vital for developing reliable LLM-based multi-agent systems.

Therefore, we summarize three research questions (RQs) to systematically study the Mandela effect in LLM-based multi-agent systems:

- **RQ₁** – Does the Mandela effect occur in LLM-based multi-agent systems?
- **RQ₂** – What factors influence the emergence of the Mandela effect?
- **RQ₃** – How can we effectively mitigate the Mandela effect?

To address **RQ₁**, we introduce MANBENCH, a novel Mandela effect-oriented benchmark featuring four typical types of tasks susceptible to the Mandela effect. It includes 20 tasks with a total of 4,838 questions and five interaction protocols. These protocols are designed to probe agent behavior in different types of social influence and memory timescales. We evaluate 13 representative LLMs, encompassing both commercial and open-source models. Moreover, we design multiple metrics to measure the Mandela effect, including error rate, reality shift rate, and maximal reality shift rate. For **RQ₂**, we investigate the key factors influencing the Mandela effect, including agent group composition, group size, knowledge domain, model scale, and memory timescales. Finally, to answer **RQ₃**, we test two types of mitigation strategies: prompt-level defenses (*e.g.*, cognitive anchoring and source scrutiny) and a model-level alignment approach to validate their effectiveness in reducing false memories. Our main contributions are summarized as follows.

- We introduce MANBENCH, a benchmark specifically designed to evaluate the Mandela effect in LLM-based multi-agent systems. MANBENCH provides systematic testing across diverse tasks and interaction protocols to probe language-driven memory bias.
- With the proposed MANBENCH, we present a comprehensive study on LLM-based collaborative multi-agent systems, measuring the impact of the Mandela effect through quantitative metrics. We also provide a detailed analysis of the factors influencing the phenomenon.
- We propose both prompt-level strategies and model-level alignment as defenses to mitigate the Mandela effect and discuss the implications of our findings for future research in LLM ethics and collaborative multi-agent systems.

3 MANBENCH

In this section, we introduce MANBENCH, a benchmark designed to evaluate the Mandela effect in multi-agent environments. MANBENCH consist of three main parts: (i) a curated set of tasks susceptible to the Mandela effect with 4,838 questions (Section 3.1); (ii) a suite of five interaction protocols that simulate various social influence scenarios to cause such effects (Section 3.2); and (iii) a set of rigorous evaluation metrics to quantitatively measure the Mandela effect (Section 3.3).

3.1 TASK CURATION AND CLASSIFICATION

To address **RQ₁**, we curate tasks from BIG-Bench Hard (BBH) (Srivastava et al., 2023) to construct ManBench as they align with the two core components of the Mandela effect. First, their verifiable ground truth provides the necessary anchor to measure memory deviation. Second, the questions under each task with multiple-choice questions that include plausible distractors create ambiguity, making agents more susceptible to influence from plausible but incorrect social content. For each question, we prompt LLM to select the most plausible incorrect answer as a **distractor** (see Appendix C.4.3 for details). We strategically classify these curated tasks into four domains to dissect the phenomenon from multiple angles: (i) **History, Time, & Events**, which highlights discrepancies in history, and gives the Mandela effect its name; (ii) **Misconceptions & Social Cognition**, which probe agents’ cognitive vulnerabilities to contagious, widespread misconceptions; (iii) **General Knowledge**, establishes the verifiable ground truth for assessing whether an agent’s memory deviates from reality; and (iv) **Domain-Specific Knowledge**, which evaluates whether these cognitive vulnerabilities extend to specialized domains. This categorization transforms general tasks into purpose-built tasks to evaluate the Mandela effect. Our final dataset comprises 4,838 multiple-choice questions, after subsampling (Turpin et al., 2023), with further statistics provided in Appendix C.1.

3.2 INTERACTION PROTOCOLS

With tasks established, to answer **RQ₂**, we develop five interaction protocols as shown in Figure 2: a baseline protocol to establish factual reality and four protocols to implant collective false memories, simulating the Mandela effect and exploring factors influencing it.

Phase 1: Anchoring Baseline Reality. The *Baseline Reality Protocol* (Figure 2a) involves questioning a subject agent in isolation to verify its baseline knowledge of a given fact. This assessment establishes an uninfluenced knowledge baseline. Building on this, we develop four interaction protocols to simulate various social scenarios. This step ensures that any memory deviations observed later are due to social interactions, rather than pre-existing cognitive flaws.

Phase 2: Implanting Collective False Memory. Based on the baseline, we introduce four additional interaction protocols to simulate the Mandela effect. Agents are subjected to one of four protocols designed to produce convincing false narratives targeting the distractor identified in Section 3.1 to implant collective false memories. As shown in Table 1, these scenarios vary by *group composition* (the cause of the effect) and *memory timescale* (the persistence of the effect). The prompts used to generate responses for different agents in each protocol are detailed in Appendix C.4.

(i) Group Composition.

A key factor behind the Mandela effect is social influence that forms and strengthens false memories. We compare two types

of social influences. The *Generic Group* simulates simple social consensus, where undifferentiated agents take turns talking without assigned roles, providing specious evidence to reinforce the consensus. In contrast, the *Role-based Group* uses a more advanced, narrative-driven interaction with five specialized agents who deliver complementary and strategically distinct evidence—each tailored to a specific role—to construct a multi-faceted false reality. We design five specialized agents: the *Error Conclusion Initiator*, who initially presents the false answer; the *Detail Support Provider*, who adds fabricated yet plausible details; the *Group Consensus Reinforcer*, who creates social proof by agreeing with the specious evidence to give the illusion of majority consensus; the *Authority Endorser*, who acts as an expert, using academic jargon to legitimize false memories; and the *Questioning Compromiser*, who first expresses doubt but is eventually convinced by the group’s narrative.

(ii) Memory Timescale. It distinguishes between an immediate, situational response to social influence and a durable, internalized false memory through memory storage and retrieval. In *Short-term* mode, the agent’s response is assessed immediately within the same conversational context as the social influence. This measures the direct, in-context impact of the interaction without any intervening memory process. In contrast, the *Long-term* mode simulates the complete lifecycle of memory formation and recall. This process involves two stages that mirror human cognition. First, a memory consolidation step occurs, where the key conclusions from the dialogue are distilled into a concise summary of beliefs. This process helps the agent form a lasting memory of the event. Second, the memory retrieval step is tested by providing this summary back to the agent as contextual background in a new interaction, forcing it to rely on its own recalled memory rather than the original, detailed dialogue. The combination of these two axes yields four distinct protocols:

- *Generic Short-term Protocol* (GS, Figure 2b). This protocol combines a Generic group with a Short-term memory timescale. The subject agent faces false peer consensus before responding to test its vulnerability to specious evidence, simulating social contagion behind the Mandela effect.
- *Generic Long-term Protocol* (GL, Figure 2c). This protocol also uses the Generic group source but shifts to a Long-term memory timescale. After initial exposure to the false consensus, the agent is re-queried alone, relying on its consolidated memory to see if the collective fallacy has become a stable, individual memory.
- *Role-based Short-term Protocol* (RS, Figure 2d). This protocol uses a Role-based group in a Short-term memory timescale. It tests if a well-crafted narrative from a specialized roles can more effectively create a false belief with high credibility than generic group.
- *Role-based Long-term Protocol* (RL, Figure 2e). This protocol uses Role-based group in a Long-term memory timescale. After memory consolidation, the agent is re-queried to measure if the strategically implanted false memory has become a deeply rooted conviction.

3.3 EVALUATION METRICS

We define a set of evaluation metrics to quantify the Mandela effect’s existence (**RQ₁**), the impact of factors (**RQ₂**), and the effectiveness of mitigation strategies (**RQ₃**). First, we establish our formal notation. Let $\mathcal{Q}$ be the set of all questions in the dataset. We track the subject agent’s response on $\mathcal{Q}$ under a specific protocol P (e.g., baseline reality protocol is represented by “B”). $\mathcal{Q}_\checkmark^P$ and $\mathcal{Q}_\times^P$ refer to the correctly answered and wrongly answered questions under specific protocol P , respectively. Across questions $\mathcal{Q}$ under the protocol P , we use the following metrics to evaluate the Mandela effect: (i) **Error rate** (Err^P), which measures wrongly answered questions. (ii) **Reality shift rate** (σ^P), which represents the proportion of questions that the agent answered correctly in the baseline reality protocol but answered incorrectly after group interaction in protocol P , measuring the shift from a correct memory. It is defined as the proportion of an agent’s correct original memories overwritten by the false collective memory in a given protocol P , with the formula:

$$\text{Err}^P = |\mathcal{Q}_\times^P|/|\mathcal{Q}|, \quad \sigma^P = |\mathcal{Q}_\times^P \cap \mathcal{Q}_\checkmark^B|/|\mathcal{Q}_\checkmark^B|. \quad (1)$$

To precisely distinguish between experimental conditions, we introduce a superscript for the group composition (G for Generic Group, R for Role-based Group) and a subscript for the memory timescale (S for Short-term memory, L for Long-term memory). Thus, the shift rates for our four protocols are denoted as σ^{GS} , σ^{GL} , σ^{RS} , and σ^{RL} , respectively. (iii) **Maximal reality shift rate** (σ_{max}). To provide a single, high-level metric for overall model comparison. This metric quantifies the total proportion of an agent’s correct baseline memories that are compromised by at least one of the four social protocols, thereby capturing the full scope of its vulnerability. It is defined as:

$$\sigma_{max} = |(\mathcal{Q}_\times^{GS} \cup \mathcal{Q}_\times^{GL} \cup \mathcal{Q}_\times^{RS} \cup \mathcal{Q}_\times^{RL}) \cap \mathcal{Q}_\checkmark^B|/|\mathcal{Q}_\checkmark^B|. \quad (2)$$

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

To address **RQ₁**, we evaluate 13 representative LLMs on MANBENCH using the five interaction protocols to quantify the Mandela effect across different protocols and model architectures. Our evaluation covers 7 commercial models (GPT-4o-mini (OpenAI, 2024a), GPT-4o (OpenAI, 2024b), GPT-5 (OpenAI, 2025), Claude 3.5 Haiku (Anthropic, 2024), Claude 4 Sonnet (Anthropic, 2025), Gemini 2.5 Flash (Team et al., 2025), and Gemini 2.5 Pro (Comanici et al., 2025)) and 6 open-source LLMs (Llama3 series (Meta, 2024; 2025), Deepseek-V3.1 (Liu et al., 2024), and Qwen3 (Yang et al., 2025) series). Detailed model settings of all LLMs are given in Appendix C.2.

4.2 MAIN RESULTS

All evaluated LLMs are susceptible to the Mandela effect. Table 2 displays their error rates Err^P under different protocols P . While models exhibit varying initial knowledge levels, with models like GPT-5 showing a stronger initial understanding of facts (17.63% error rate under the Baseline Reality Protocol) compared to others like Llama3.1-8B, which has the highest error rate (44.58%). Despite these differences in capability, all models exhibit a significant increase in error rate when subjected to social influence. For instance, Qwen-235B’s error rate rises from a baseline

of 25.48% to 74.75% under the Role-based Short-term Protocol. Even the best-performing GPT-5 is not immune, with its error rate more than doubling to 41.59% under the Role-based Short-term Protocol. These findings indicate that no current LLM is fully resistant to the social construction of false realities, regardless of its initial knowledge capabilities.

Short-term false memories can solidify into long-term beliefs. Table 3 shows the reality shift rate σ^P across four protocols. Models such as GPT-5 and Llama3.3-70B show short-term susceptibility but strong long-term memory integrity, with GPT-5’s reality shift rate dropping from 31.03% (σ^{RS}) to just 1.67% (σ^{RL}). They have strong self-correction, preventing false memories from becoming stable beliefs. Conversely, models like Claude 3.5 Haiku, Llama3.1-8B and Qwen3 series internalize false memories into long-term beliefs. For instance, Claude 3.5 Haiku’s shift rate remains high from 63.67% (σ^{RS}) to 55.63% (σ^{RL}). Notably, Llama3.1-8B’s rate rises from 61.69% (σ^{GS}) to 85.13% (σ^{GL}). These findings reveal that these models solidify false memories into stable beliefs, increasing the risks of misinformation.

4.3 WHAT DRIVES THE MANDELA EFFECT?

To answer **RQ₂**, we section examines the factors that influence the Mandela effect. Specifically, we analyze the impact of the following aspects: the agent group composition (Section 4.3.1), the memory timescale (Section 4.3.2), the agent group size (Section 4.3.3), the epistemic properties of the knowledge domain (Section 4.3.4), and the model scale of agents (Section 4.3.5).

4.3.1 GROUP COMPOSITION

Role-based group is more potent at inducing the Mandela effect than Generic Group across most models. Table 3 shows that for nearly all LLMs, the reality shift rate under role-based protocols (σ^{RS}, σ^{RL}) is higher than under generic protocols (σ^{GS}, σ^{GL}). This difference is more pronounced in advanced models. For example, Claude 4 Sonnet’s reality shift rate rises from 15.45% in the Generic Short-term Protocol to 35.21% in the Role-based Short-term Protocol. Similarly, GPT-4o’s reality shift rate increases from 46.04% (σ^{GS}) to 55.95% (σ^{RS}). This indicates that narrative complexity and perceived credibility amplify the Mandela effect. The only exception is Deepseek-V3.1, which is more vulnerable to Generic Group ($\sigma^{GS} = 60.60\%$) than Role-based Group ($\sigma^{RS} = 47.81\%$), suggesting a unique cognitive bias toward simple consensus over narrative complexity.

4.3.2 MEMORY TIMESCALE

The Mandela effect decreases in long-term memory due to memory decay. As shown in Table 3, most LLMs exhibit lower reality shift rates in long-term memory (σ^{GL}, σ^{RL}) compared to short-term memory (σ^{GS}, σ^{RS}). Advanced models like GPT-5 and Llama3.3-70B view the Mandela effect as an in-context illusion that fails to solidify into durable beliefs. For example, GPT-5’s reality shift rate drops from 31.03% (σ^{RS}) to 1.67% (σ^{RL}). However, models like Claude 3.5 Haiku and Qwen3 retain false memories persistently, with Claude 3.5 Haiku retaining most of its initial false beliefs, as its reality shift rate only slightly drops from 63.67% (σ^{RS}) to 55.63% (σ^{RL}). These models consolidate false narratives into long-term memories, which poses a greater risk of misinformation.

4.3.3 GROUP SIZE

Figure 3 shows how group size influences the Mandela effect under different protocols. Compared to the Generic Group, the Role-based Group can detect the effect when the number of agents exceeds a threshold. Detailed role compositions and interaction sequences for different group sizes are provided in Appendix C.3.

In the Generic Group, the Mandela effect intensifies as agents increase and saturate at a critical group size.

Under generic protocols (GS and GL), both the error and reality shift rates rise with more agents before plateauing. For example, the reality shift rate of GPT-4o-mini and Claude 3.5 Haiku increases from about 37% (σ^B) with one agent to roughly 56% (σ^{RL}) when the group reaches seven members under the Generic Short-term Protocol, indicating that seven agents are sufficient to exert maximum influence.

In the Role-based Group, the Mandela effect first increases and then decreases as the number of agents increases.

The inverted-U curve observed under the Role-based protocol shows both the error rate and reality shift rate initially rise sharply, peaking at a group size of six agents (e.g., σ^{RS} exceeding 63%). After this point, the Mandela effect diminishes with error rates and reality shift rates dropping. For groups of nine or more agents, the overall error rate falls below the Baseline Reality Protocol (e.g., Err^{RS} and Err^{RL} both under 32%). We believe this is due to a “suspicion-induced vigilance” effect: while a small group of experts seems credible, a large, coordinated group of agents is viewed as a suspicious conspiracy and triggers increased critical thinking, leading the agent to self-correct some of its initial errors and perform better than it would alone.

Key observations. The curve of the Generic Group’s influence demonstrates that a system’s factual integrity can be compromised by a surprisingly small number of agents. The inverted-U curve of the Role-based Group reveals that the greatest risk comes not from the largest possible group, but rather from moderately-sized, strategically coordinated groups that project high credibility without arousing suspicion. Conversely, **this “suspicion-induced vigilance” effect suggests agents possess a latent capability to detect inauthentic social dynamics**, a finding that provides the direct theoretical basis for the targeted, prompt-level interventions we propose and validate in Section 5. Our source scrutiny defense proactively uses this mechanism. By explicitly prompting the model to analyze narratives and evaluate credibility, this defense activates the vigilance mechanism without requiring a large group size, enabling the model to effectively identify and reject false memories.

4.3.4 KNOWLEDGE DOMAIN

As shown in Table 4, our evaluation covers four knowledge domains. Our analysis reveals a vulnerability: the Mandela effect is not limited to narrative and ambiguous topics but also occurs strongly in domains based on factual and specialized knowledge.

Mandela effect thrives in narrative and ambiguity domains. Domains such as “History, Time, & Events” and “Misconceptions & Social Cognition” are highly susceptible to the Mandela effect, exhibiting substantial reality shift rates (σ^{RS}) of 58.74% and 52.67%, respectively. These narrative-driven domains have fragile memories with ambiguities. This provides a perfect entry point for carefully crafted false memories. Consequently, this domain exemplifies the classic manifestation of the phenomenon, where a convincing narrative easily exploits existing memory gaps.

Even areas with strong baseline knowledge are vulnerable to the Mandela effect, especially in specialized domains. For “General Knowledge”, despite a low baseline error rate of 9.40%, we observe that the reality shift rates (σ^{GS}) of 48.06%, providing decisive evidence that social influence can systematically overwrite correct memories. The “Domain-Specific Knowledge” is even more vulnerable, with the highest initial shift among all categories at 67.46% (σ^{RS}) and the most persistent long-term false belief ($\sigma^{RL}=37.77\%$). This suggests the Mandela effect mainly risks corrupting established knowledge in high-stakes, specialized areas.

4.3.5 MODEL SCALE

Simply scaling up model size does not necessarily reduce the Mandela effect. Figure 4 shows the maximal reality shift rate σ_{max} across models of different sizes. For some model families, scaling is an effective mitigation strategy. This is most evident in the Claude 3.5 family, where Claude 3.5 σ_{max} decreases from 72.0% (Haiku) to 39.6% (Sonnet), with the GPT family showing similar improvements. Others, like Qwen3, show an inverse scaling law, with σ_{max} rising from 89.3% (8B) to 92.2% (235B), suggesting larger models might be more vulnerable to the Mandela effect. This suggests that the Mandela effect isn’t simply a knowledge deficiency that can be solved by more parameters. Instead, larger models may be better at understanding complex false narratives, but not necessarily lead to improved critical thinking. Their superior narrative understanding may make them more susceptible to internalizing false narratives, increasing their susceptibility to the Mandela effect.

5 MITIGATION STRATEGIES

As mentioned in Section 4.3.3, agents have a latent ability to detect inauthentic social dynamics, which motivates us to explore methods to mitigate the Mandela effect. To address **RQ₃**, we first develop two prompt-level defenses (Section 5.1) and then propose a model-level defense (Section 5.2) as preliminary defenses to strengthen the models’ ability to resist the Mandela effect.

5.1 PROMPT-LEVEL DEFENSE

Our findings indicate that the Mandela effect is driven by distinct social mechanisms, primarily the social contagion of false memories and the implantation of a compelling false narrative. With this in mind, we propose two prompt-based interventions: *cognitive anchoring* and *source scrutiny*. These strategies guide the agent from passively accepting collective memory to actively verifying facts. The prompts used for prompt-level defense are available in Appendix D.1.

Cognitive anchoring. This “inside-out” defense counters social consensus through three principles: (i) *Primacy of internal knowledge* requires the agent to establish a “cognitive anchor” and base conclusions solely on its own knowledge, isolated from social influence. (ii) *Skepticism towards external claims* involves critically examining the group’s shared memory against this anchor and identifying discrepancies. (iii) *Burden of proof for belief change* demands that the agent justify any deviation from its initial anchor. These principles transform the agent’s cognitive process from passive acceptance to active evaluation, strengthening resistance to collective false memories.

Source scrutiny. This “outside-in” defense resists false narratives by shifting the agent’s role from passive to a critical analyst of discourse. It relies on three principles: (i) *Presumption of Influence* emphasizes conversational dynamics and persuasive intent rather than surface claims. (ii) *Narrative Deconstruction* identifies strategic roles and rhetorical patterns. (iii) *Credibility as an Output* bases judgments on structural analysis, viewing unnatural or overly coherent consensus as manipulation. These principles help the agent analyze narratives and neutralize inauthentic collective realities.

Results. As shown in Figure 5, we observe two findings about defenses against the Mandela effect. (i) **Prompt-level defenses effectively disrupt the social contagion of false evidence.** Both cognitive anchoring and source scrutiny dramatically reduce the reality shift rate compared to the undefended baseline. For instance, when faced with Generic Short-term Protocol (σ^{GS}), GPT-4o’s susceptibility was slashed from a baseline of 46.0% to 17.8% by cognitive anchoring and 26.5% by Source Scrutiny. This shows that a reorganized reasoning process effectively counters the development of a collective false memory. (ii) **Cognitive anchoring more effectively reduces short-term influence, but both defenses similarly prevent long-term belief solidification.** Under short-term protocols, cognitive anchoring outperforms source scrutiny. For GPT-4o, cognitive anchoring slashes the reality shift rate σ^{RS} from 56.0% to 17.0%, while source scrutiny achieves 25.2%. Under long-term protocols, their effectiveness converges. For GPT-4o, both strategies reduce σ^{RL} from 33.6% to a comparably low level (15.2% for cognitive anchoring and 14.5% for source scrutiny). This suggests that both strategies prevent false memory from becoming durable convictions.

5.2 MODEL-LEVEL DEFENSE

While prompt-level defenses provide external reasoning support, a more robust approach is to embed resilience as an intrinsic model capability. We implement a model-level defense using Supervised Fine-Tuning (SFT) (Wu et al., 2025b). The goal of SFT is to help the model resist manipulative false narratives without excluding valid guidance from other agents. This involves fine-tuning LLMs on a balanced dataset that includes a resilience set to resist manipulative social influence and a cooperative set to accept valid guidance. The details for constructing these sets, including the prompts used to generate them, composition ratios, and training hyperparameters, are provided in Appendix D.2.

Resilience set. This subset is designed to train the agent to develop cognitive resilience against false narratives from other agents. It includes reasoning chains generated by applying our cognitive anchoring and source scrutiny prompts in Section 5.1. We select reasoning chains that successfully defend against the Mandela effect to form the resilience set.

Cooperative set. A SFT dataset composed solely of the resilience set poses a critical risk: the model may not learn to critically evaluate social context, but instead to categorically exclude it, making it challenging to adaptively select useful contexts and filter irrelevant ones. To solve this problem, the cooperative set is introduced to teach the model how to accept truths in a productive manner. This subset is composed of two specific types of cooperative scenarios: (i) **Corrective guidance.** Cases where the agent is initially wrong and the group provides the correct answer, with the ideal response demonstrating belief updating. (ii) **Enriching guidance.** Cases where the agent is initially correct and the group provides valuable supporting details, with the ideal response showing constructive integration. To determine whether the agent will unconditionally filter all replies from other agents, we introduce a new Correct Guidance Protocol C , where agents share a correct narrative and calculate σ^C as the proportion of questions answered correctly in the baseline but incorrectly after group interaction in protocol C .

Results. As shown in Figure 6, our fine-tuning experiments reveal three critical findings on the mitigation of the Mandela effect. (i) **Agent responses to collective memories can be shaped through supervised fine-tuning.** Models trained on our resilience set learn to effectively defend their factual memories against social contagion (e.g., Llama3’s σ^{RS} drops from 99.5% to 18.2%), while those trained on the cooperative set learn to update their beliefs when presented with correct guidance from other agents, showing a reality shift rate under Correct Guidance Protocol (σ^C) of 0.8%. (ii) **Training only on the resilience set makes the model unconditionally exclude other agents’ answers.** Fine-tuning solely on the resilience set causes the model to dogmatically dismiss all social input. This is evident under the Correct Guidance Protocol, where the agent’s σ^C surges to 38.5% (for Llama3) when all other agents provide correct memory, even worse than its baseline model. (iii) **Mitigating the Mandela effect requires balanced training for discernment.** The model trained on the combined dataset reduces susceptibility to false memories (e.g., Llama3’s σ^{RS} drops to 21.5%) and maintains the ability to learn from valid social input (σ^C at 1.1%). This shows that true cognitive resilience necessitates training an agent to distinguish between manipulative and helpful contexts. A truly robust agent must know both when to be skeptical and when to learn.

7 CONCLUSION

In this paper, we find that the Mandela effect—the collective false memories of a verifiable fact—is a significant and emerging vulnerability in LLM-driven multi-agent systems. Through our proposed benchmark, MANBENCH, we systematically investigate the existence, persistence, and quantitatively measure this phenomenon. We find that nearly all LLM-based agents are highly vulnerable to the Mandela effect, especially in role-based narratives, with the greatest susceptibility in areas of ambiguity and specialized knowledge. Furthermore, we propose prompt-level and model-level defenses to mitigate this effect. We hope our findings and the MANBENCH framework will encourage the development of more resilient and reliable LLM-based multi-agent systems.